

Traian Florin Șerbănuță

Associate Professor · Researcher in Formal Methods · Software Engineer

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Education

PhD in Computer Science

2010

University of Illinois, Urbana-Champaign

Dissertation: *A Rewriting Approach to Concurrent Programming Language Design and Semantics*

Advisor: Grigore Roșu

Committee: Thomas Ball, Darko Marinov, José Meseguer, Madhusudan Parthasarathy

Master in Computer Science

2004

University of Bucharest

Dissertation: *Institutional Concepts in First-Order Logic, Parameterized Specification Theory, and Logic Programming*

Advisors: Răzvan Diaconescu, Virgil Emil Căzănescu

Bachelor in Computer Science

2002

University of Bucharest

Dissertation: *Information Hiding in Text Using LR(k) Grammars*

Advisor: Adrian Atanasiu

Experience

Associate Professor of Computer Science

2013–Present

University of Bucharest, Faculty of Mathematics and Informatics

Courses developed and taught, with all materials openly published:

- Software Systems Modelling — requirements analysis and modelling (UML, design patterns)
- Declarative Programming — functional and declarative programming in Haskell
- Concurrency in Programming Languages — concurrency hands-on across Java, C++, Erlang/Elixir, JavaScript, and Python
- Programming Languages Semantics — operational semantics, interpreters, and type systems
- Foundations of Programming Languages — hands-on semantics, lambda calculus, and type systems in Haskell and Prolog, with an introduction to logic programming
- Program Verification — Hoare logic, weakest preconditions, separation logic, SAT/SMT solvers, symbolic execution, and model checking
- Introduction to Machine Learning — hands-on machine learning for non-computer-scientists (Master in Digital Humanities; Kaggle notebook)

Co-founder and Vice-President

2022–Present

Institute for Logic and Data Science

Non-profit research institute (ilds.ro) bringing together academic and industry researchers in logic and data science, with an active program of conferences, workshops, weekly seminars, and hosted research projects. Most involved in the **Working Formal Methods Symposium** (FROM 2024, 2026), the *Deep Blockchain Fundamentals* workshop, the weekly **Logic** seminar, and matching-logic research.

Consultant and Researcher

2026–Present

Asymptotic

Prototyping AI-assisted, Lean-based provers for programming languages. Lead author of *Rust-Prover*, a Lean 4-backed verifier for Rust: specifications and Rust code are translated into Lean theorems that LLM agents prove, with Lean's kernel as the final check. It proved all 1325 theorems of the 1007-problem VeriContest benchmark.

Consultant and Researcher

2024–2026

Pi Squared, Inc.

Core Rust engineer on Pi Squared's verifiable-computing and universal-settlement infrastructure ("Proof of Proof"). Work spans five phases of the platform's evolution:

- **Pi² research prototype (2024)** — Metamath proof checkers compiled to five zkVM backends (RISC Zero, SP1, Nexus, Lurk, Delphinus), with driver tooling and cross-backend benchmarking.
- **Blocks to Circom ZK pipeline (2025)** — Rust toolchain compiling the Blocks DSL into Circom subcircuits and generating end-to-end ZK certificates for block instantiations.
- **Verifiable Settlement Layer (2025)** — claim-based settlement layer where submitted claims reach quorum and are recorded as signed settlements.
- **FastSet validator and proxy (2025–2026)** — validator and client-facing proxy for the FastSet protocol.
- **Fast Shop / agentic AI commerce (2026)** — Universal Commerce Protocol integration with region-aware multi-region delivery, a Shopify backend, and an MCP-exposed commerce surface.

Consultant and Researcher

2012–2023

Runtime Verification, Inc.

Long-running principal contributor across four projects:

- **K Framework — Java implementation (2012–2015)** — executor/debugger internals for the Maude and Java backends: LTL model-checking, search-graph extraction, and AC matching.
- **RV-Predict — race & deadlock detector (2012–2017)** — maximal-causal-model predictive race detection for Java and C/C++ (via an LLVM instrumentation pass), scaled to a 1M-variable cap.
- **K Haskell Backend — symbolic-execution prover (2017–2022)** — all-path/one-path reachability-logic prover with SMT integration, applied to the formal semantics of the EVM, WebAssembly, and IELE.
- **CBC-Casper / VLSM consensus in Coq (2022–2023)** — Coq modelling and verification of blockchain protocols, including one of the first safety formalizations of CBC Casper.

Postdoctoral Research Fellow

2011–2013

Alexandru Ioan Cuza University, Iași (FMSE Laboratory)

Formal methods research in software engineering. Coordinated the team developing the K Framework.

Postdoctoral Research Associate

2011–2012

University of Illinois, Urbana-Champaign (Information Trust Institute)

Formal systems and verification research.

Research Assistant

2004–2010

University of Illinois, Urbana-Champaign (FSL Laboratory)

Assisted in research on formal semantics, rewriting logic, and programming language design. Designed and prototyped (in Maude) the K semantic framework.

Summer Intern

2007

Google, New York

Co-authored a patent application on web traffic analysis methods (with Bogdan Căpriță).

Summer Intern

2005

Microsoft Research, Redmond (Testing, Verification, and Measurement Group)

Contributed an equality theory propagation core for the Zap automated theorem prover.

Teaching Assistant

2003–2004

University of Bucharest, Department of Computer Science Fundamentals

Supported undergraduate courses in programming, discrete mathematics, and computer science theory.

Programmer

2000–2001

Popnet-Agentscape Romania, Natural Language Processing Team

Implemented classification algorithms for one of the first AI agents.

Open-source Projects

Personal research and tooling outside employer-affiliated work; full list at <https://github.com/traiansf>.

- **Formalization libraries** — aml-in-coq (Applicative Matching Logic), arl-in-coq (Abstract Rewrite Systems), sets-in-coq.
- **Type theory study** — propositions-as-types, Coq scribbles along *Type Theory and Formal Proofs* (Nederpelt & Geuvers).
- **Teaching companions** — semantics-in-coq and semantics-in-lean, companions to the Foundations of Programming Languages course.
- **Teaching tools** — multiple_choice_exam, built with AI coding agents: a Python generator for randomized, print-ready exam variants from a Markdown question bank, and a Dart mobile app that grades answer sheets by QR code and optical mark recognition.
- **Other** — excel-database, a WordPress plugin (★4) that exposes an Excel spreadsheet table as queryable data.

Skills

Formal methods and verification Programming language semantics AI-assisted software development
LLM agents for theorem proving Rewriting logic K Framework Temporal logics Model checking
Specification and verification techniques

Programming Languages

Rust Haskell Coq / Rocq Java C / C++ K / Maude Python Lean 4 Prolog
JavaScript / TypeScript PHP

Languages

Romanian (native) · English (fluent) · French (basic)

Publications

55 articles · h-index 22 · 2178 citations (Google Scholar, as of October 2026)

Top 20 by citations (Google Scholar, October 2026). Full record: Google Scholar · DBLP.

1. Roşu, Grigore and Traian Florin Şerbănuţă. “An Overview of the K Semantic Framework.” *Journal of Logic and Algebraic Programming*, Vol. 79, No. 6, pp. 397-434, 2010. [673]
2. Chen, Feng, Traian Florin Şerbănuţă, and Grigore Roşu. “jPredictor: a predictive runtime analysis tool for Java.” *ICSE '08: Proceedings of the 30th International Conference on Software Engineering*, pp. 221-230, 2008. [155]
3. Şerbănuţă, Traian Florin, Grigore Roşu, and José Meseguer. “A Rewriting Logic Approach to Operational Semantics.” *Information and Computation*, Vol. 207, No. 2, pp. 305-340, 2009. [129]
4. Ştefănescu, Andrei, Ştefan Ciobăcă, Radu Mereuta, Brandon M Moore, Traian Florin Şerbănuţă, and Grigore Roşu. “All-Path Reachability Logic.” *Logical Methods in Computer Science*, Vol. 15, Issue 2, 2019. [117]
5. Luo, Qingzhou, Yi Zhang, Choonghwan Lee, Dongyun Jin, Patrick O’Neil Meredith, Traian Florin Şerbănuţă, and Grigore Roşu. “RV-Monitor: Efficient Parametric Runtime Verification with Simultaneous Properties.” *Runtime Verification (RV’14)*, LNCS Vol. 8734, pp. 285-300, 2014. [108]
6. Şerbănuţă, Traian Florin, Feng Chen, and Grigore Roşu. “Maximal Causal Models for Sequentially Consistent Systems.” *Runtime Verification (RV’12)*, LNCS Vol. 7687, pp. 136-150, 2013. [105]
7. Şerbănuţă, Traian Florin and Grigore Roşu. “K-Maude: A Rewriting Based Tool for Semantics of Programming Languages.” *Rewriting Logic and Its Applications (WRLA’10)*, LNCS Vol. 6381, pp. 104-122, 2010. [74]
8. Şerbănuţă, Traian Florin. “Extending Parikh matrices.” *Theoretical Computer Science*, Vol. 310, No. 1-3, pp. 233-246, 2004. [66]
9. Roşu, Grigore and Traian Florin Şerbănuţă. “K Overview and SIMPLE Case Study.” *Proceedings of K’11*,

- ENTCS Vol. 304, pp. 3-56, 2014. [64]
10. Roşu, Grigore, Wolfram Schulte, and Traian Florin Şerbănuţă. "Runtime Verification of C Memory Safety." *Runtime Verification (RV'09)*, LNCS Vol. 5779, pp. 132-151, 2009. [64]
 11. Şerbănuţă, Virgil Nicolae and Traian Florin Şerbănuţă. "Injectivity of the Parikh matrix mappings revisited." *Fundamenta Informaticae*, Vol. 73, No. 1-2, pp. 265-283, 2006. [63]
 12. Şerbănuţă, Traian Florin, Andrei Arusoai, David Lazar, Chucky Ellison, Dorel Lucanu, and Grigore Roşu. "The K Primer (version 3.3)." *Proceedings of K'11*, ENTCS Vol. 304, pp. 57-80, 2014. [54]
 13. Kasampalis, Theodoros, Dwight Guth, Brandon Moore, Traian Florin Şerbănuţă, Yi Zhang, Daniele Filaretti, Virgil Şerbănuţă, Ralph Johnson, and Grigore Roşu. "IELE: A Rigorously Designed Language and Tool Ecosystem for the Blockchain." *Formal Methods (FM'19)*, pp. 593-610, 2019. [46]
 14. Şerbănuţă, Traian Florin and Grigore Roşu. "Computationally Equivalent Elimination of Conditions." *Rewriting Techniques and Applications (RTA'06)*, LNCS Vol. 4098, pp. 19-34, 2006. [41]
 15. Rusu, Vlad, Dorel Lucanu, Traian-Florin Şerbănuţă, Andrei Arusoai, Andrei Ştefănescu, and Grigore Roşu. "Language Definitions as Rewrite Theories." *Journal of Logical and Algebraic Methods in Programming*, Vol. 85, No. 1, pp. 98-120, 2016. [34]
 16. Daian, Philip, Ylies Falcone, Patrick Meredith, Traian Florin Şerbănuţă, Akihito Iwai, Shin'ichi Shiriashi, and Grigore Roşu. "RV-Android: Efficient Parametric Android Runtime Verification, a Brief Tutorial." *Runtime Verification (RV'15)*, LNCS Vol. 9333, pp. 342-357, 2015. [31]
 17. Hills, Mark, Traian Florin Şerbănuţă, and Grigore Roşu. "A Rewrite Framework for Language Definitions and for Generation of Efficient Interpreters." *Rewriting Logic and Its Applications (WRLA'06)*, ENTCS Vol. 176, No. 4, pp. 215-231, 2007. [31]
 18. Lucanu, Dorel, Traian Florin Şerbănuţă, and Grigore Roşu. "K Framework Distilled." *Rewriting Logic and Its Applications (WRLA'12)*, LNCS Vol. 7571, pp. 31-53, 2012. [29]
 19. Ellison, Chucky, Traian Florin Şerbănuţă, and Grigore Roşu. "A Rewriting Logic Approach to Type Inference." *Recent Trends in Algebraic Development Techniques (WADT'08)*, LNCS Vol. 5486, pp. 135-151, 2009. [28]
 20. Şerbănuţă, Traian Florin, Gheorghe Ştefănescu, and Grigore Roşu. "Defining and Executing P Systems with Structured Data in K." *Membrane Computing (WMC'08)*, LNCS Vol. 5391, pp. 374-393, 2009. [28]